

## Time series comparisons:

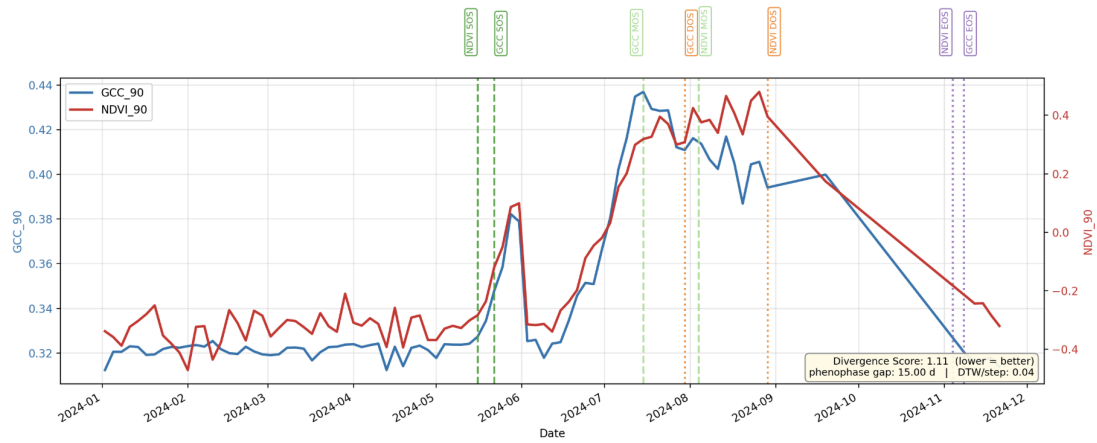
I did not end up using normal Correlation Variance (CV) metrics. CV only measures how spread out a single dataset is, it can't mathematically compare two different time series. And it seems like basic cross correlation is going to fail to capture if NDVI and GCC have the exact same seasonal shape but are just shifted a few weeks apart in timing (lag), or if the duration of the season is compressed.

Instead, I use phenophase dates and dynamic time warping (DTW). This seems to be the industry standard used by researchers like Richardson. [creator of PhenoCam] ([DTW info here](#)). It will measure whether two curves match in both biological timing and overall shape.

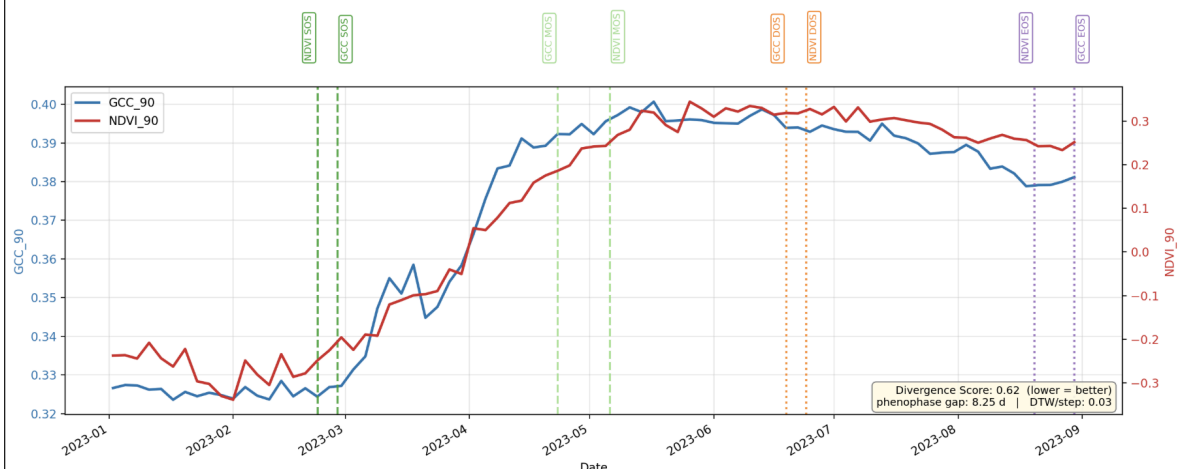
Here is how I calculated according to the article link above:

- **Phenophase Gap** (check phenophase)
  - 1st: SOS, MOS, DOS, and EOS is calculated with previous piecewise logistic functions (return null when it can't split the year into two fittable halves around a single dominant peak...usually when there is <10 points, 5 samples of either end)
  - 2nd: calculates the absolute difference in days between NDVI and GCC for each phase, and averages them together:
    - $\text{phenophase\_gap} = \text{mean}(|\text{SOS\_ndvi} - \text{SOS\_gcc}|, |\text{MOS\_ndvi} - \text{MOS\_gcc}|, |\text{DOS\_ndvi} - \text{DOS\_gcc}|, |\text{EOS\_ndvi} - \text{EOS\_gcc}|)$
    - If the SOS difference is 6 days, MOS is 20, DOS is 30, and EOS is 4, the mean gap is 15 days. A lower score means the sensors agree on the biological timing
  -
- **DTW per Step** (check the shape)
  - The general idea behind DTW is that it finds the optimal nonlinear alignment between the two curves and sums the distance between the two points along that path. (after it finds the best match, adds up all the little leftover gaps between the two curves ) Small total = the curves look almost the same. Big total = the curves really differ in shape.
  - DTW distance grow larger as the amount of day increase, so divide by the total days which turn into an average gap per day score
  -
- Overall: Divergence score
  - **divergence\_score** = (**phenophase\_gap** / CANDIDATE\_GAP\_DAYS) + **dtw\_per\_step**
  - Currently, "CANDIDATE\_GAP\_DAYS" = 14 (*let me know if you would like to change this threshold*)
    - 14 days is the phenophase gap threshold for now (tolerance for gap)
    - Ex:  $15/14 + 0.04 = 1.11$  (this is a pretty high divergence score)
- Examples plots:

API: PhenoCam Time Series for GCC\_90 and NDVI\_90 (aafcottawacfiat14n\_AG\_1000, 2024)



API: PhenoCam Time Series for GCC\_90 and NDVI\_90 (NEON.D08.TOMB.DP1.20002\_DB\_2000, 2023)



API: PhenoCam Time Series for GCC\_90 and NDVI\_90 (utwente1\_GR\_1000, 2023)



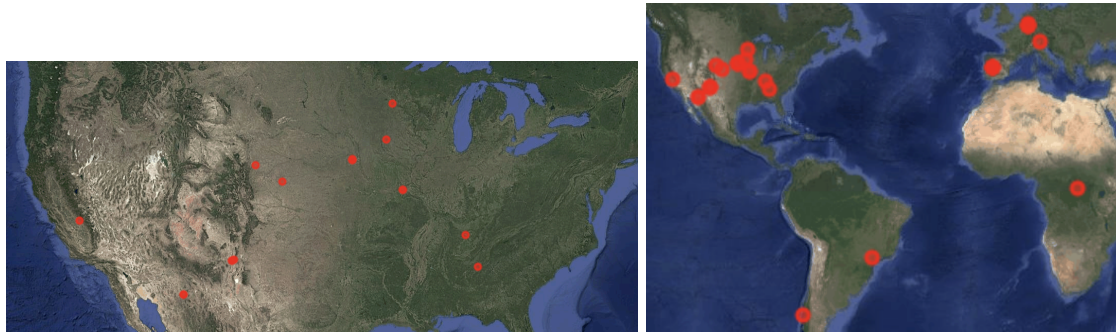
## Google Earth Engine Visualizations:



- Layer 2, NDVI: check the average vegetation greenness across the year. (trying to figure out if this landscape is patchy or homogeneous)
- Also have other layers but did not use it to perform any math yet (*let me know if you would like me to*)
  - Layer 3: GVF, how much of each pixel is vegetation
  - Layer 4: Spatial CV (coefficient of variation over time), pixel by pixel difference in NDVI over the year.
  - Layer 5a : water
  - Layer 5b : bare soil
  - Layer 5c : Urban
- Other keys:
  - Orange box: VIIRS/satellite 4km
  - Red circle: PhenoCam 200m field of view
  - White dot: camera location
- Ex: Layer 5b + Layer 2



Finally, Lat and Lon of sites that I find uniform and NDVI & GCC has strong agreements:

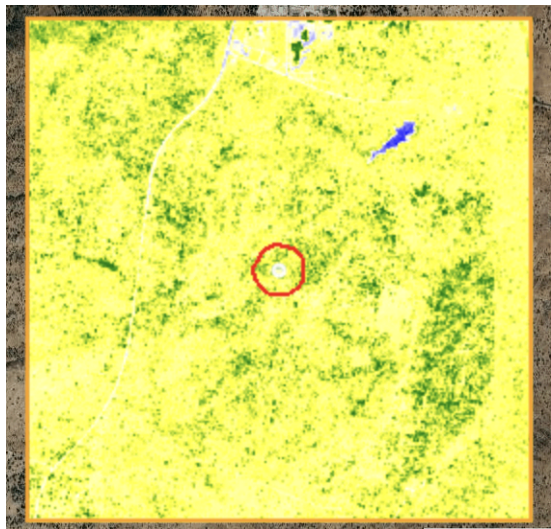


First find the top 40 sites that have the lowest divergence score, then find Coefficient of variation (stdDev/mean) for both phenocam and satellite of least or equal to 0.4, then delete any duplicate/ very close site. **2023/2024**

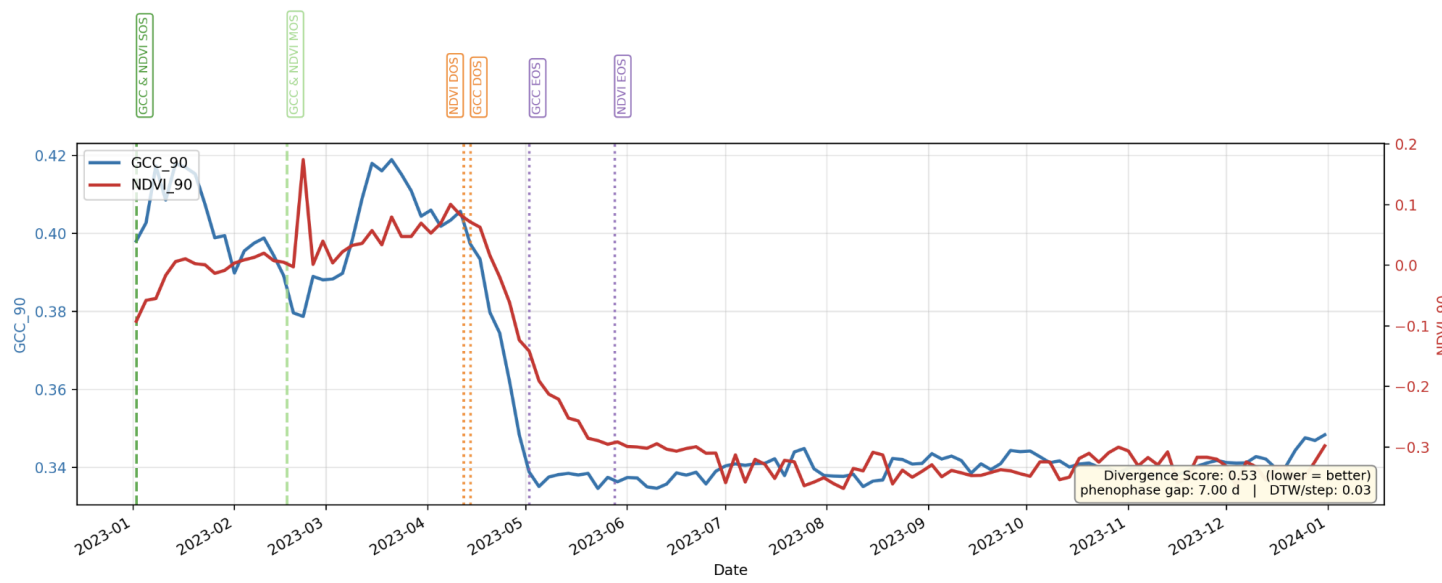
- 1: (lat: 39.9426889 lon: -5.7786833) eslm1\_EB\_1002 **23**
- 2: (lat: 34.3386 lon: -106.739) sevmvecreo22ambinc\_SH\_1000 //USA **23**
- 3: (lat: 39.22848 lon: -92.11936 ) goodwater\_AG\_1000 //USA **23**
- 4: (lat: 33.8875 lon: -85.6935) piedmontalpasture2\_GR\_1000 //USA **23**
- 5: (lat: 50.625 lon: 6.9844 ) breedface2024control2\_AG\_100 **24**
- 6: (lat: 36.1171 lon: -86.8295) tsubiology2\_GR\_1000 //USA **23**
- 7: (lat: -40.1726 lon: -73.4439 ) alercecosteroforest\_EN\_1001 **23**
- 8: (lat: 31.91068 lon: -110.83549) NEON.D14.SRER.DP1.00042\_SH\_1000 //USA **24**
- 9: (lat: 39.9426889 lon: -5.7786833) eslm1\_EB\_1000 **24**
- 10: (lat: 41.1797 lon: -96.4397) mead3\_AG\_1000 //USA **23**
- 11: (lat: 42.4816 lon: -93.5235 ) arscleessouth\_AG\_1000 //USA **23**
- 12: (lat: 39.758246 lon: -102.447103) NEON.D10.ARIK.DP1.20002\_GR\_1000 //USA **24**
- 13: (lat: 44.6946 lon: -93.0578) rosemountcons\_AG\_2000 //USA **23**
- 14: (lat: 46.6767 lon: 10.5773) it25matschb1p\_EN\_1000 **24**
- 15: (lat: 50.8632 lon: 6.5311) apvcontrol\_AG\_1000 **23**
- 16: (lat: 0.8144 lon: 24.5024) congoflux\_EB\_1000 **23**
- 17: (lat: -22.967875 lon: -48.728009) eucflux\_EB\_1000 **23**
- 18: (lat: 50.6249 lon: 6.9871) breedface2024control1\_AG\_1000 **23**
- 19: (lat: 37.10878 lon: -119.73228) NEON.D17.SJER.DP1.00042\_GR\_1000 //USA **23**
- 20: (lat: 40.8055764 lon: -104.7558542) cperuvb\_GR\_1000 //USA **23**

Example:

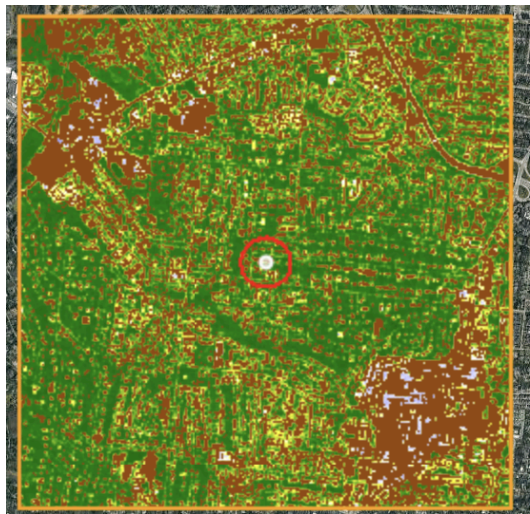
Site 19:



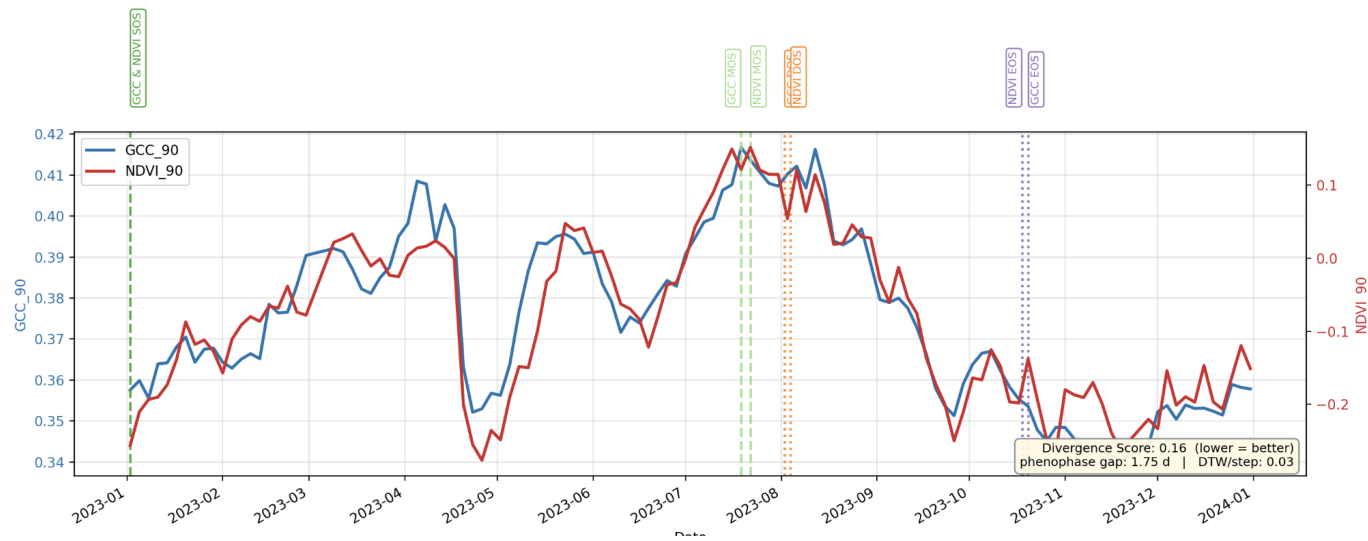
API: PhenoCam Time Series for GCC\_90 and NDVI\_90 (NEON.D17.SJER.DP1.00042\_GR\_1000, 2023)



Site 6:

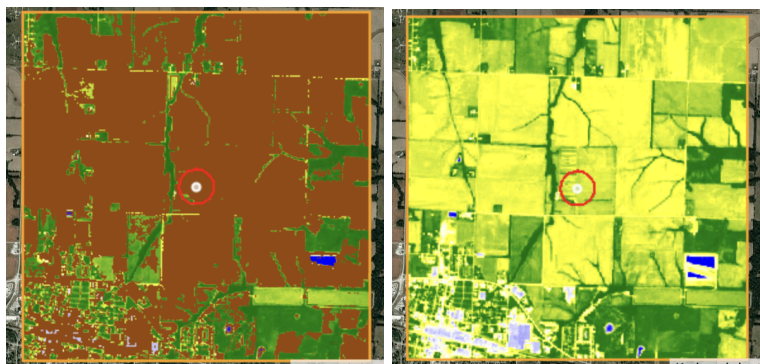


API: PhenoCam Time Series for GCC\_90 and NDVI\_90 (tsubiology2\_GR\_1000, 2023)





Site 3:



API: PhenoCam Time Series for GCC\_90 and NDVI\_90 (goodwater\_AG\_1000, 2023)

